CHAPTER 35

Assessment & Certification

Instructor Deck — DevSecOps Lab Book Knowledge Assessment

- Self-Assessment Quizzes
- 200+ Question Exam Bank
- 4 Hands-On Lab Exercises
- 3-Tier Certification Path

Session Agenda

How this chapter's materials map to instruction time



1. Self-Assessment Quizzes

5 diagnostic questions covering Phase 1–3 fundamentals (DNS, IAM, Kubernetes, Vault).



2. Certification Exam Bank

200+ questions across 3 levels: Foundation, Intermediate, Advanced.



3. Hands-On Lab Exercises

4 timed labs (30–90 min) validating practical, applied skills.



4. Certification Levels

Practitioner → Engineer → Architect career progression and requirements.

Self-Assessment Quiz — Phase 1-3 Knowledge Check

Instructor note: reveal answers after class discussion — correct answer marked in gold



Q1. What is the primary advantage of split-brain DNS?

A) Faster responses

B) Different responses for internal/external queries

C) Reduced cost

D) Easier configuration



Q2. What does OIDC provide that LDAP does not?

A) Password management

B) User directories

C) Single sign-on with modern protocols

D) Group management



Q3. Which component stores all Kubernetes cluster state?

A) API Server

B) Scheduler

C) etcd

D) Controller Manager

Self-Assessment Quiz — Continued

Discuss why distractors are tempting — this reveals common misconceptions



Q4. What is the purpose of a SecurityContext in Kubernetes?

A) Encrypt network traffic

B) Enforce pod security policies

C) Manage RBAC rules

D) Create secrets



Q5. In Vault, what is a "lease"?

A) A time-bound permission

B) A secret storage location

C) An encryption algorithm

D) A backup procedure

Certification Exam Bank — 200+ Questions

Three progressive difficulty levels, each mapping to real operational maturity

LEVEL 1: FOUNDATION

50

questions

- DNS & Networking — 5
- Identity & Auth — 10
- HTTPS/TLS — 5
- Kubernetes Basics — 15
- Secrets Mgmt — 10
- Access Control — 5

LEVEL 2: INTERMEDIATE

75

questions

- CI/CD Pipelines — 15
- Container Security — 15
- Infrastructure as Code — 15
- Compliance & Auditing — 15
- Incident Response — 15

LEVEL 3: ADVANCED

75

questions

- High Availability — 15
- Disaster Recovery — 15
- Multi-Region — 15
- Network Security — 15
- DevSecOps Culture — 15

Hands-On Lab Exercises

Applied validation exercises — schedule as guided labs or timed checkpoints

Exercise 1: Deploy DNS

30 min

Objective: Configure BIND9 split-brain DNS
Validation: Dig queries return correct results
Files: Chapter 02 reference configs

Exercise 2: Kubernetes Hardening

45 min

Objective: Apply pod security policies
Validation: Pods cannot escalate privileges
Files: Chapters 23–29 security contexts

Exercise 3: Incident Response

60 min

Objective: Respond to simulated incident
Validation: Timeline reconstructed correctly
Files: Chapter 31 forensics procedures

Exercise 4: Compliance Audit

90 min

Objective: Collect compliance evidence
Validation: All controls demonstrated
Files: Chapter 32 evidence checklists

Certification Pathway

Career progression from foundational knowledge to strategic architecture



LEVEL 1

DevSecOps Practitioner

PREREQUISITES

Basic Linux, networking

EXAM

50 questions / 60 min

PASSING SCORE

Passing: 70%

VALUE

Foundation understanding



LEVEL 2

DevSecOps Engineer

PREREQUISITES

Level 1 + 6 months experience

EXAM

75 questions / 120 min

PASSING SCORE

Passing: 75%

VALUE

Operational capability



LEVEL 3

DevSecOps Architect

PREREQUISITES

Level 2 + 2 years experience

EXAM

75 questions + capstone

PASSING SCORE

Passing: 80%

VALUE

Design & strategy

Instructor Delivery Notes

Pacing

Run the 5-question quiz as a warm-up (~10 min); use it to gauge readiness before assigning labs.

Lab Sequencing

Schedule labs in order — DNS → Kubernetes → Incident Response → Compliance builds complexity progressively (~3.75 hrs total).

Exam Alignment

Map each exam domain back to its source chapter so learners can self-remediate gaps before sitting Level 1–3 exams.

Certification Cadence

Recommend a minimum 6-month gap between Level 1 and Level 2 to accumulate real operational experience.



Ready to Assess & Certify

Quizzes • 200+ Question Exam Bank • Hands-On Labs • 3-Tier Certification